

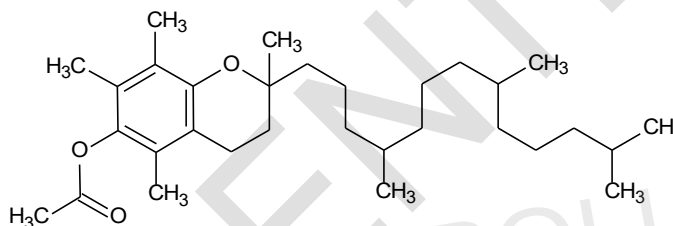
Product Data Sheet

Appearance: Almost white or yellowish powder or granular powders

Active ingredient

Chemical name	3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl acetate; 2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-chroman-6-yl acetate
EINECS No.	231-710-0
CAS No.	7695-91-2
Synonyms	All rac- α -Tocopheryl acetate; dl- α -Tocopheryl acetate; Vitamin E acetate; Vitamin E

Structural formula



Molecular mass 472.75 g/mol

Molecular formula $C_{31}H_{52}O_3$

Specification

Description	Vitamin E 50% is manufactured by mixing oily Vitamin E with inorganic carrier.
Content(Vitamin E)	Min 50.0% (equal to 500.000IU/kg) (European Pharmacopoeia EP-0691)
Bulk density	In between 550 g/L and 650 g/L
Loss on drying	Max 5.0%
Lead	Max 2mg/kg

Other chemical and physical data

Particle size	Min. 99% pass through a sieve of 0.85 mm
Solubility	Insoluble in water

Packaging and storage conditions

Packaging	Plastic big bag (500 kg net), Paper bag (25 kg net) and others on request.
Storage	Store in a dry and cool place.

Stability

The product can be kept for 24 months in the unopened, original packaging at room temperature. Once opened, use content soonest (best used up not more than one month). Expiry date: see label resp. certificate of analysis.

Uses and EC-regulations

Product to be used for premixes and feeding stuffs for all animal species and categories. Use in liquid feed (e.g. milk replacer) is not recommended. Product is approved by regulation EU 2023/341 as feed additive (approval number 3a700i) and fulfills all requirements of EC Regulation 1831/2003 for feed additives resp. EC Directive 2002/32 regarding undesired substances in feed additives. For handling of product resp. information on safety please see ZMC Material Safety Data Sheet.

Legal notice

The information given herein is believed to be correct and is in accordance with our current state of knowledge. No warranty is made, either expressed or implied, regarding its accuracy or the results to be obtained from the use of such information. Information given does not relieve any user from own reliability for safe handling under his conditions of use and in accordance with all applicable laws and regulations in force.

SAFETY DATA SHEET

1. Identification of the substance /mixture and of the company/undertaking

Product Identifier Vitamin E 50%, feed grade

Synonym: All rac- α -Tocopheryl Acetate; dl- α -Tocopheryl Acetate; Vitamin E acetate; Vitamin E

Product Code: 41000014

Recommended Use: Additive for animal nutrition in premixes and compound feeds

Manufacturer Information:

Company Name Zhejiang Medicine Co., Ltd. Changhai Biological Company

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2. Hazards identification

Hazard Classification

This product is not classified as a hazardous substance or as dangerous goods according to the classification

2.2 Other hazards

The product is under certain conditions capable of dust explosion.

3. Composition/information on ingredients

Chemical Name:	CAS No.	Concentration (w/w)*
Vitamin E Acetate	7695-91-2	Min 50%
Silicon Dioxide	7631-86-9	Max 50%

Remarks: *The data is based on average.

4. First-aid measures

Description of first aid measures

Inhalation: Remove from area of exposure and move to fresh air. Consult physician immediately.

Skin contact:	Flush exposed area with plenty of water and soap. Seek medical advice if irritation persists.
Eye contact:	Flush eyes with plenty of water for at least 15 minutes. Seek medical advice if irritation persists.
Ingestion:	Rinse mouth immediately. Do not induce vomiting unless directed to do so by medical personnel. Consult physician immediately.

Most important symptoms or effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in 2 and/or in 11. Further important symptoms and effects are so far not known.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-fighting measures**Extinguishing media**

Suitable Extinguishing media: Water spray, foam, carbon dioxide (CO₂), dry powder.

Unsuitable Extinguishing media: None known.

Special hazards arising from the substance or mixture

Dust explosion hazard: fine dust dispersed in air in sufficient concentrations and in the presence of an ignition source is a potential dust explosion hazard.

Advice for fire-fighters

Wear full fire-fighting turn-out gear (full Bunker gear) including NIOSH-approved self-contained breathing apparatus.

6. Accidental release measures**Personal precautions**

Avoid dust formation. Do not breathe dust. Ensure adequate ventilation. Use personal protective apparatus. Information regarding personal protective measures see 8.

Environment precautions: Do not discharge into drains/surface or water courses.

Methods and material for containment and cleaning up

Sweep/shovel up or vacuum up. Collect in a suitable container for disposal.

7. Handling and storage

Precautions for safe handling	- Avoid dust formation, provide for appropriate exhaust ventilation
	- Take precautionary measures against electrostatic charging;
Conditions for safe storage	- Store in a tightly closed container in a dry and cool place at the room temperature.

8. Exposure controls/personal protection

Occupational exposure limits

Information on carrier Silicon Dioxide

Maximum airborne concentration; max 4 mg per cubic meter at the workplace for inhalable dust.

Nuisance dust: 8 hours TWA = 10 mg/cubic meter

Personal protective measures

Respiratory protection:	Suitable respiratory protection like half-mask.
Hand protection:	Glove material: for example nitrile rubber.
Eye protection:	Safety glasses with side-shields (frame goggles)
Body protection:	Protective suit and Hygiene measures.

General safety and hygiene measures

When using, do not eat or drink. Remove and wash contaminated clothing before re-use. Ensure adequate ventilation, especially in confined areas.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance:	Almost white or yellowish
Form:	Powder or granular powders
Odor:	Almost odorless
Odor threshold:	No data available
pH:	Not applicable
Melting point/range:	84.5°C (184.1°F)
Boiling point/range:	Decomposes.
Flash point:	Not applicable
Evaporation rate:	Not applicable
Flammability (solid/gas):	May form combustible dust concentrations in air.
Vapour pressure:	Not applicable
Vapour density:	Not applicable
Bulk density:	0.55~0.65 g/cm ³
Solubility:	Insoluble in water
Partition coefficient: n-octanol/water:	No data available
Auto-ignition temperature:	Not applicable
Thermal Decomposition:	No data available
Viscosity:	Not applicable.

Explosion parameters

Maximum explosion pressure, P_{max} /MPa	0.48
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Maximum rate of pressure rise (dp/dt), max(MPa/s)	3.4
Dust specific measure of explosibility, $K_{st}(Mpa \cdot m/s)$	0.93
Lowest explosion level, $LEL(g/cm^3)$	50-100
Min Ignition temperature C (dust cloud), $MITC(^{\circ}C)$	> 400
Min Ignition temperature C (dust layer), $MITL(^{\circ}C)$	> 400
The minimum ignition energy, MIE/mJ	
Product particle size	250-500
Sieved sample, 100% <0.10mm	50-100

The MIE of a powder strongly depends on the particle size distribution and the water content of the tested sample: the finer and the dryer the powder the lower the MIE.

10. Stability and Reactivity

10.1 Reactivity:	No hazards to be specially mentioned.
10.2 Chemical stability:	Stable under recommended storage conditions.
10.3 Possibility of hazardous reactions:	Dust may form explosive mixture in air.
10.4 Conditions to avoid:	Avoid all sources of ignition: heat, sparks, open flame, generating dust, moisture, static discharges and light.
10.5 Incompatible materials:	Incompatible with oxidizing agents, acid chlorides, acid anhydrides.
10.6 Hazardous combustion products:	No hazardous decomposition if used as directed.

11. Toxicological information

Information on toxicological effects

<u>Acute toxicity:</u>	Fish: LC_{50} (96h) > 100mg/l ; NOEC (96h) > 100mg/l Daphia: EC_{50} (48h) > 100mg/l NOEC (96h) > 100mg/l LD 50 (rat): 5000 mg/kg
<u>Skin corrosion/ Irritation:</u>	Contact with dust can cause mechanical irritation or drying of the skin. Prolonged skin contact may cause skin irritation.
<u>Eye damage/ irritation:</u>	Dust contact with the eyes can lead to mechanical irritation.
<u>Sensitisation to the respiratory tract/skin:</u>	Non-sensitizing.
Carcinogenicity:	Not classified
Mutagenicity:	Not classified.
Reproductive toxicity:	Not classified.
STOT- single exposure:	Not classified.

STOT-repeated exposure: Not classified.

12. Ecological information

Ecotoxicity: No data available
Persistence and degradability: Not readily biodegradable.
17 % (28d)(OECD Test Guideline 301F).
Bioaccumulation potential: No data available.
Mobility in soil: No data available.
Results of PBT & vPvB assessment: No data available.
Other adverse effects: No data available.

13. Disposal considerations

Dispose of contents/container in accordance with local/regional/national/international regulations.

14. Transport information

Land transport

Not classified as a dangerous good under transport regulations

Sea transport

Not classified as a dangerous good under transport regulations

Air transport

Not classified as a dangerous good under transport regulations

15. Regulation Information

Regulation Information In according with Regulation (EC) No 1272/2008 and 29 CFR part 1910.1200

16. Other information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. This information is provided without any warranty, express or implied, regarding its correctness. The conditions or methods of handling, storage, use or disposal of the product are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of the product.